| | | | |
|---|---|---|---|
| **Fiscal Year:** | FY 2016 | **Task Last Updated:** | FY 04/26/2016 |
| **PI Name:** | Myers, Jerry Ph.D. | | |
| **Project Title:** | Metric Development to Quantify Change in Mission Risk Due to a Physician-Trained Crew Medical Officer | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Operational and clinical research | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **ExMC**:Exploration Medical Capabilities | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | Jerry.G.Myers@nasa.gov | **Fax:** | FY |
| **PI Organization Type:** | NASA CENTER | **Phone:** | 216-433-2864 |
| **Organization Name:** | NASA Glenn Research Center | | |
| **PI Address 1:** | Human Research Office MS 110-3 | | |
| **PI Address 2:** | 21000 Brookpark Road | | |
| **PI Web Page:** | | | |
| **City:** | Cleveland | **State:** | OH |
| **Zip Code:** | 44135 | **Congressional District:** | 10 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | Directed Research |
| **Start Date:** | 06/01/2015 | **End Date:** | 12/31/2015 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Antonsen, Erik | **Contact Phone:** | 281.483.4961 |
| **Contact Email:** | erik.l.antonsen@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | N/A | | |
| **COI Name (Institution):** | George, Richard ( Summa Health Systems )<br>McCarroll, Michele ( Summa Health Systems ) | | |
| **Grant/Contract No.:** | Directed Research | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | NASA-STD-3001 currently requires a physician-trained Crew Medical Officer (CMO) for missions exceeding 210 days in length. NASA roughly based this standard on the nominal 180-day length of an International Space Station (ISS) mission, which affords relatively quick and easy return to Earth for treating complicated cases, with an assumption that longer missions would not have such an option and thus would need to operate more autonomously. As NASA considers various Design Reference Missions (DRMs) for exploration, however, there are DRMs of only a few weeks duration that could not quickly return to Earth. Accordingly, they may well carry a medical risk level high enough to warrant the judgment of a physician-trained CMO. In contrast, the current assumption that missions to Mars would be less medically risky with a physician-trained CMO may be invalid. Due to rigorous preflight screening, which lessens the likelihood of a disease state, and the limited nature of the envisioned medical kits, perhaps NASA's treatment options are limited enough that the judgment of a physician-trained CMO would not appreciably reduce medical mission risk. | | |

| | |
|---|---|
| **Task Description:** | The point of the preceding discussion is that NASA does not have an objective metric on which to base the decision of which missions require a physician CMO. Hence, Human Research Program Element Exploration Medical Capabilities (ExMC) Gap 2.02, which states: We do not know how the inclusion of a physician crew medical officer quantitatively impacts medical risk during exploration missions. Developing a quantified assessment of the change in risk as a function of mission type will first require developing a metric by which clinical judgment can be assessed. Currently the terrestrial medical community does not have a consensus in this area of medical education. NASA cannot afford to wait for such a consensus because of the need to provide such analyses for mission concept trade studies. If such an analysis were available today, NASA could use that information. Therefore, in order to close Gap 2.02, we first focus on an unanswered question in medical education: how does one assess clinical judgment? With the development, validation, and implementation of a metric representing clinical judgment, ExMC can use the Integrated Medical Model (IMM) to quantify the change in risk posture as a function of mission type associated with having a physician-trained CMO.<br><br>Accordingly, the task described in this Task Synopsis consists of a pilot study to develop and validate such a metric, to be completed by Summa Health Systems with advice from the IMM team about the suitability of the metric for modeling. After this pilot study comes to an end and the Task is completed, it is recommended to ExMC that the pilot study be repeated at another institution to validate it in an analog population in order to correct for an organizational bias, and to increase subject numbers in order to improve statistical significance. A larger study could also be completed to gather statistically significant data for probabilistic risk assessment application if needed by the Element to successfully quantify the change in risk associated with a physician CMO. .<br><br>Aims:<br><br>Specific Aim 1: Development. To develop a Medical Judgment Pathway Metric (MJPM) for assessing clinical judgment based on evidenced-based practice for treating two common conditions from the Exploration Medical Condition List (EMCL):<br><br>o Infection ; o Chest Pain/Pneumothorax/Angina<br><br>Specific Aim 2: Validation. To validate the MJPM by a multi-disciplinary expert medical team to confirm principal components of the MJPM. For this wave of expert review, each randomly ordered item is followed by a response table. The experts will read each statement and indicate under which domain(s) they believe the symptom response best fits as far as clinical judgment (e.g., pathway 1, pathway 2, or pathway 3).<br><br>Specific Aim 3: Analog Testing. To test the MJPM in an analog population using populations that mirror those of the astronaut corps in terms of both age and training (n=20: 12-males and 8-females) of<br><br>o Group 1: Board Certified physician in an administrative position (n=10: 6-males and 4-females)<br><br>o Group 2: Masters or Doctoral prepared in a specific science field (n=10: 6-males and 4-females). |
| **Rationale for HRP Directed Research:** | 1. Insufficient time for solicitation: The metric to be developed in this task will feed the development of the dynamic Probabilistic Risk Assessment (PRA) module which is part of the Integrated Medical Model (IMM) v5.0. The team is currently scheduled to complete the dPRA architecture by 2/2016; if a pilot validated metric is not available by then, the model architecture may not support using dPRA to quantify the effect of a physician crew medical officer on mission risk. The directed study can meet that deadline; solicitation cannot.<br>2. Highly constrained research: Not enough information is known in order to solicit given the cutting edge nature of this work. This task is a pilot study and follow-on tasks for solicitation are recommended.<br><br>3. Highly constrained research: The development of metrics in this task requires working with the IMM team, which is very specific to NASA, to develop inputs for the dPRA module. A portion of the metric needs to be developed in parallel with the input design for the dPRA module. |
| **Research Impact/Earth Benefits:** | Work supports clinical faculty with additional evaluation metrics, outside of typical skill-based metrics, to assess training progress in medical students and resident physicians on Earth. The MJPM provides a promising tool to assess medical decision-making in general. |
| **Task Progress:** | During the project period the projected achieved two aims; 1. Develop a Medical Judgment Pathway Metric (MJPM) assessing clinical judgment; and 2. Demonstrate evidence of validity and reliability of MJPM scores. The individual MJPM items, domains, and sections of the MJPM were built based on existing standardized frameworks. Content validity was determined by a convenient sample of eight experts. The MJPM instrument was pilot tested in four medical simulations with a team of three medical raters assessing 40 participants with four levels of medical experience and skill.<br>Results: Raters were highly consistent in their MJPM scores in each scenario (ICC 0.965 to 0.987) as well as their evaluation of the expected patient outcome (Fleiss's Kappa 0.791 to 0.906). For each simulation scenario, average rater score cut-scores significantly predicted expected loss of life or stabilization (Cohen's Kappa 0.851 to 0.880).<br><br>Discussion: The MJPM demonstrated preliminary evidence of reliability and validity. |
| **Bibliography Type:** | Description: (Last Updated: 06/18/2025) |
| **Articles in Peer-reviewed Journals** | Ahmed RA, McCarroll ML, Schwartz A, Gothard MD, Atkinson SS, Hughes PG, Cepeda Brito JR, Assad L, Myers JG, George RL. "Development, validation, and implementation of a medical judgment metric." MDM Policy & Practice. 2017 Jan;2(1):2381468317715262. https://doi.org/10.1177/2381468317715262 [NOTE reported originally in 4/2016 as "In press as of April 2016."] , Jan-2017 |